

Q1.) Hierarchical Clustering Technique

Hierarchical clustering is an **unsupervised machine learning technique** used to group similar data points into clusters based on their similarity. It builds a **hierarchy (tree-like structure)** of clusters called a **dendrogram**.

1. Definition

Hierarchical clustering is a method of cluster analysis which seeks to build a hierarchy of clusters either by:

- **Agglomerative approach (Bottom-Up)**
- **Divisive approach (Top-Down)**

2. Types of Hierarchical Clustering

a) Agglomerative Hierarchical Clustering

- Starts with each data point as an individual cluster.
- Gradually merges the closest clusters step by step.
- Continues until all points form a single cluster.

□ Also called **Bottom-Up approach**

b) Divisive Hierarchical Clustering

- Starts with all data points in one cluster.
- Recursively splits into smaller clusters.
- Continues until each point is a separate cluster.

□ Also called **Top-Down approach**

3. Key Concepts

1. Distance Measures

Used to calculate similarity:

- Euclidean Distance
- Manhattan Distance
- Cosine Similarity

2. Linkage Criteria

Defines how distance between clusters is measured:

- **Single Linkage** → Minimum distance
- **Complete Linkage** → Maximum distance
- **Average Linkage** → Average distance
- **Centroid Linkage** → Distance between centroids

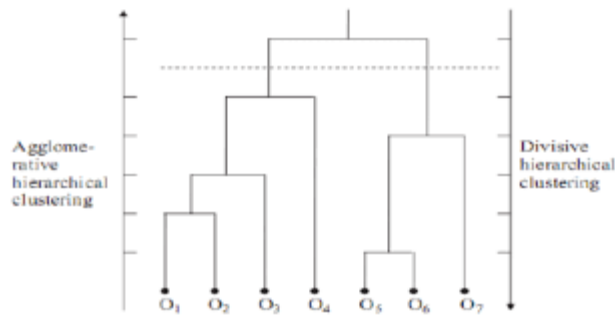
4. Algorithm (Agglomerative Approach)

Steps:

1. Start with **N clusters (each data point is a cluster)**
2. Compute the **distance matrix**
3. Find the **two closest clusters**
4. Merge them into one cluster
5. Update the distance matrix
6. Repeat until only one cluster remains

5. Dendrogram Representation

A **dendrogram** is a tree diagram that shows how clusters are formed.



- X-axis → Data points
- Y-axis → Distance (similarity level)
- Cutting the dendrogram at a certain height gives clusters

6. Example of Hierarchical Clustering

Given Data Points:

A(1,1), B(2,1), C(4,3), D(5,4)

Step-by-Step:

1. Calculate distances between all points
2. Closest points: A & B → Merge → Cluster (AB)
3. Next closest: C & D → Merge → Cluster (CD)
4. Now merge (AB) and (CD) at higher distance

Final Clusters:

- Cluster 1: A, B
- Cluster 2: C, D

□ This process is clearly visible in the dendrogram.

7. Advantages

- No need to specify number of clusters beforehand
- Produces a **hierarchical structure**
- Easy to interpret using dendrogram
- Works well for small datasets

10. Comparison (Agglomerative vs Divisive)

Feature	Agglomerative	Divisive
Approach	Bottom-Up	Top-Down
Start	Individual points	Single cluster
Complexity	Less complex	More complex
Usage	More common	Rare

Q2.) Expectation–Maximization (EM) is a **probabilistic clustering technique** used to find clusters by assuming that the data is generated from a mixture of probability distributions (commonly **Gaussian distributions**).

It is widely used in **Gaussian Mixture Models (GMM)**.

1. Definition

EM clustering is an **iterative algorithm** that assigns data points to clusters based on probabilities and refines cluster parameters to maximize the likelihood of the data.

□ Unlike K-Means (hard clustering), EM performs **soft clustering** (each point can belong to multiple clusters with different probabilities).

2. Key Idea

- Data is assumed to come from **K distributions**
- Each cluster is represented by:
 - Mean (μ)
 - Variance (σ^2)
 - Mixing probability (π)
- EM alternates between:
 - **Expectation (E-step)**
 - **Maximization (M-step)**

3. Steps of EM Algorithm

Step 1: Initialization

- Choose number of clusters (K)
- Initialize parameters (means, variances, probabilities)

Step 2: Expectation Step (E-Step)

- Compute probability that each data point belongs to each cluster

$$P(\text{cluster}_i | x) = \frac{P(x | \text{cluster}_i) \cdot P(\text{cluster}_i)}{\sum_{j=1}^K P(x | \text{cluster}_j) \cdot P(\text{cluster}_j)}$$

□ This gives **soft assignments**

Step 3: Maximization Step (M-Step)

- Update cluster parameters using computed probabilities:
 - Update mean
 - Update variance
 - Update mixing coefficients

Step 4: Repeat

- Repeat E-step and M-step until convergence (parameters stop changing)

4. Example

Consider a dataset of heights of people belonging to two groups:

- Group 1 → Short people
- Group 2 → Tall people

Initially, we don't know which person belongs to which group.

Iteration 1:

- Randomly assign probabilities

E-Step:

- Calculate probability of each person belonging to "short" or "tall"

M-Step:

- Update mean height and variance for both groups

After Several Iterations:

- Two Gaussian clusters are formed:
 - Cluster 1 → Short
 - Cluster 2 → Tall

□ Each person has a probability of belonging to both clusters.

5. Advantages of EM Clustering

1. **Soft Clustering**
 - Assigns probabilities instead of fixed clusters
2. **More Flexible than K-Means**
 - Can model clusters of different shapes and sizes
3. **Works with Probabilistic Models**
 - Useful in real-world uncertain data
4. **Handles Overlapping Clusters**

- Better than hard clustering methods
- 5. **Statistically Strong**
 - Based on maximum likelihood estimation

6. Limitations of EM Clustering

1. **Computationally Expensive**
 - Slower than K-Means
2. **Sensitive to Initialization**
 - Poor initialization may lead to wrong clusters
3. **May Converge to Local Optima**
 - Not guaranteed to find global optimum
4. **Requires Number of Clusters (K)**
 - Must be specified in advance
5. **Assumption of Distribution**
 - Usually assumes Gaussian distribution (may not fit all data)

7. Comparison with K-Means

Feature	EM Clustering	K-Means
Type	Soft Clustering	Hard Clustering
Model	Probabilistic	Distance-based
Shape	Any (elliptical)	Spherical
Output	Probabilities	Fixed assignment
Complexity	High	Low